

West Virginia University
Department of Economics,
Elementary Business and
Economics Statistics
Fall 2026

Class Time: Asynchronous
Class Location: Online

ECON 225-701 CRN: 11801

Instructor: Joshua Teye

Instructor email: jt00064@mix.wvu.edu

Online Office Hours: MWF 12:30 – 1:30 PM and TR By Appointment (via Zoom)

Course Description This course explores basic concepts of statistical models, distributions, probability, random variables, tests of hypotheses, confidence intervals, regression and correlation with emphasis on business and economics examples. (Equiv. to STAT 211.)

Prerequisites “C-” or better in MATH 122, 123, 126, 129, or 153 or successful completion of Math 150, 154, 155, or 156 with a minimum grade of “D-”.

Office Hours and E-Mail Communication: I will be available for online office hours via Collaborate or Zoom on all weekdays (unless otherwise stated) by appointment. I will do my best to reply to any emails within 24 hours.

Required Course Materials

Textbook: *Introductory Business Statistics*. Holmes, Illowsky, and Dean. (2017) via OpenStax. Note that this textbook is available for **free** and will be integrated into our Top Hat course.

Access to TopHat:

We will be using the Top Hat (www.tophat.com) classroom response system in class. You will be able to submit answers to in-class questions using Apple or Android smartphones and tablets, laptops, or through text message.

You can visit the Top Hat Overview (<https://success.tophat.com/s/article/Student-Top-Hat-Overview-and-Getting-Started-Guide>) within the Top Hat Success Center which outlines how you will register for a Top Hat account, as well as providing a brief overview to get you up and running on the system.

An email invitation will be sent to you by email, but if don't receive this email, you can register by simply visiting our course website: <https://app.tophat.com/e/820716>

→ **Note: our Course Join Code is 820716**

Top Hat does require a paid subscription, and a full breakdown of all subscription options available can be found here: www.tophat.com/pricing. The plan you will need for this course is the 4-month term (\$33).

Should you require assistance with Top Hat at any time, due to the fact that they require specific user information to troubleshoot these issues, please contact their Support Team directly by way of email (support@tophat.com), the in app support button, or by calling 1-888-663-5491.

Course Objectives and Expected Learning Outcomes

You should leave this course with a basic understanding regarding the relevance and usefulness of statistics (including its limitations), as well as a working knowledge of key tools used in carrying out statistical analyses. You will also learn to utilize Excel's spreadsheet program so that you have hands-on experience applying statistics to real data. This course addresses two of WVU's General Education Curriculum (GEC) objectives:

G02 (Group A): Basic Mathematical Skills and Scientific Inquiry: Use quantitative and scientific knowledge effectively.

G04: Contemporary Society: Apply knowledge, methods and principles of inquiry to contemporary problems, ideas and/or values.

GEF: Equivalently, this course addresses the new General Education Foundations (GEF) requirement F3. Mathematics & Quantitative Skills.

The expected learning outcomes for the course are the following:

1. organize and display data in a meaningful way and communicate its essential features using tables and charts;
2. compute and interpret summary measures of data, especially measures of central tendency and dispersion;
3. apply basic probability, probability distributions (especially the Standard Normal Distribution), and sampling distributions to problems and to real data;
4. conduct and interpret confidence interval estimation and hypothesis testing;
5. conduct and interpret correlation analysis and simple linear regression analysis.

Grading Policy

Your final course grade will be comprised of weekly chapter assignments on TopHat, participation, random video quizzes, and a final project. There is a total of 500 points for the course.

Assignments	Total points	Date	Location
Chapter Notes & Quizzes	150	Weekly	TopHat
Exams (2)	100	TBD	TopHat
Final Project	50	TBD	TopHat
Total points	300		

Your final grade letter is given according to the percentage of your total points earned to the total possible points in this class. Below is the scale of the letter grades:

Grade letter	Points
A	270 - 300
B	240 – 269.9
C	210 – 239.9
D	180 – 209.9
F	<180

Assignments on TopHat

The following list describes the individual components of the online homework assignments and quizzes through TopHat.

(1) Chapter Notes Assignments and Participation:

Material on TopHat will be organized into weekly folders which will contain the assigned chapter(s) to be read from the textbook, as well as notes that summarize the main terms/ideas/formulas. These notes will also include homework-style questions such as true/false, multiple choice, fill in the blank, and more that will be designed to test your understanding of the material. These questions will be scored on two dimensions: (1) correctness and (2) participation. You will have the entire week during which the material is assigned to complete these questions.

(2) Video Quizzes:

Short videos will be periodically embedded in the chapter notes on TopHat that will demonstrate examples of material from the textbook. Emphasis will be placed on material involving calculations. You will be required to watch the video and then answer questions based on the specific problem shown. Quizzes are assigned on Friday and due the same night by 11:59 PM.

(3) Exams:

Two multiple-choice exams will be administered online via TopHat and will be available for the entire day (24 hours) they are scheduled for. All answers must be submitted by 11:59 PM.

Final Project

In place of a final exam, this course will have a final project. This project will consist of multiple layers of data analysis, in which students will be required to collect, summarize, and perform various inferential statistical techniques on their own data. Work will be performed in Microsoft Excel, which is available for free to students [here](#). Further details will be provided later in the semester.

Late Work Policy

Absolutely *no* late work will be accepted unless a doctor's note or travel form can be provided that covers the entire duration of the assignment's availability. I will grant an extension on any graded material as long as you contact me **before** the assignment's due date.

Midterm Grade Calculation

It is anticipated that students will have completed approximately 50 percent of the above assignments by mid-semester and that students' midterm grade will be calculated based on approximately 50 percent of the possible points in the course.

Extra Credit

There will be a few opportunities to earn extra credit points as they arise throughout the semester. I will announce when these opportunities become available and for how many points they will be worth via e-mail (I will also post on e-Campus). You should not plan to rely on extra credit to bump you up a letter grade at the end of the semester. I will also not respond to any e-mails asking about extra credit unless I have already made an announcement.

Inclusivity Statement

The West Virginia University community is committed to creating and fostering a positive learning and working environment based on open communication, mutual respect, and inclusion. If you are a person with a disability and anticipate needing any type of accommodation to participate in this class, please advise me and make appropriate arrangements with the Office of Disability Services (293-6700). For more information on West Virginia University's Diversity, Equity, and Inclusion initiatives, please see <http://diversity.wvu.edu>.

Academic Dishonesty

The integrity of the classes offered by any academic institution solidifies the foundation of its mission and cannot be sacrificed to expediency, ignorance, or blatant fraud. Therefore, I will enforce rigorous standards of academic integrity in all aspects and assignments of this course. For the detailed policy of West Virginia University regarding the definitions of acts considered to fall under academic dishonesty and possible ensuring sanctions, please see the Student Conduct Code located at <http://studentlife.wvu.edu/studentconductcode.html>.

Institutional Policies:

Students are responsible for reviewing [policies](#) on inclusivity, academic integrity, incompletes, sale of course materials, sexual misconduct, adverse weather, as well as student evaluation of instruction, and days of special concern/religious holiday statements.